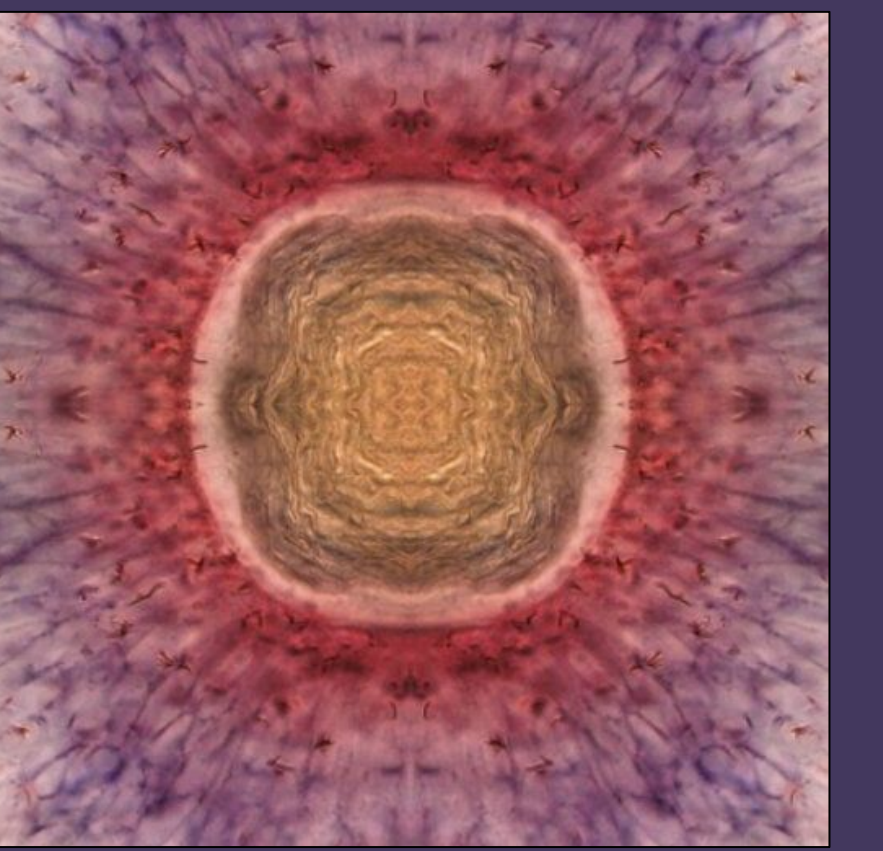




Dynamics of Cell Death in the Denervated Zebrafish Optic Tectum

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Background

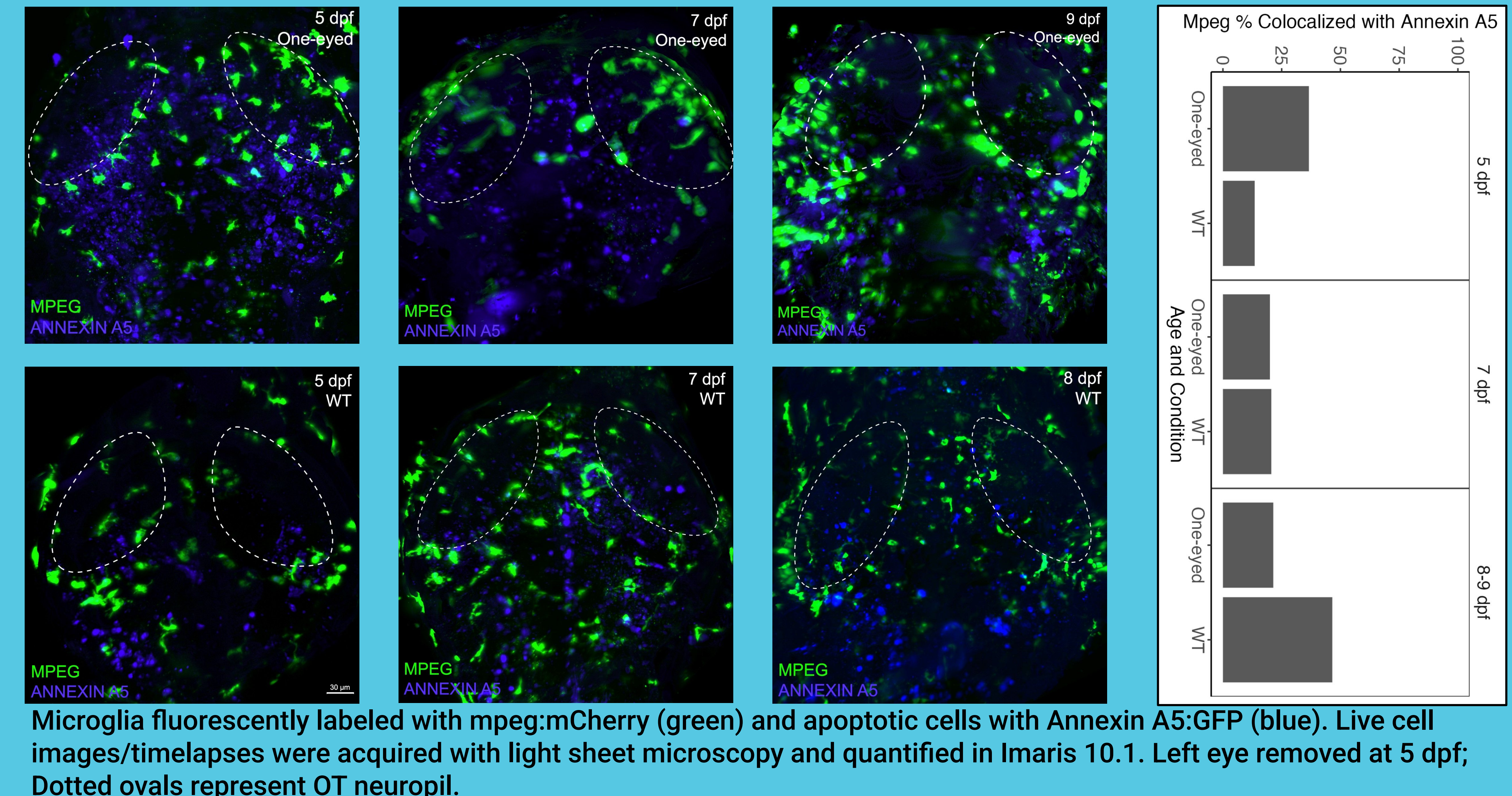
The optic nerve innervates the optic tectum (OT) and is essential for OT development and survival. Denervation leads to reduced proliferation and increased cell death. Microglia are the phagocytic cells of the CNS. Studying microglia and apoptosis dynamics as well as the interaction between these two events helps us learn about OT development, plasticity, and injury response.

Experimental Questions:

- How do microglia behaviors differ in innervated and denervated OT?
- How to study dynamic interactions across time between microglia and apoptotic cells?

Annexin A5 (apoptosis) - Mpeg

- Annexin+ cells can be seen blebbing and fragmenting
- Microglia engulf and clear clusters of apoptotic cells
- Few microglia are colocalized with Annexin A5 across groups
- Microglia morphology seems to differ between groups and ages
- Microglia are preferentially localized in the OT neuropil areas of one-eyed larvae



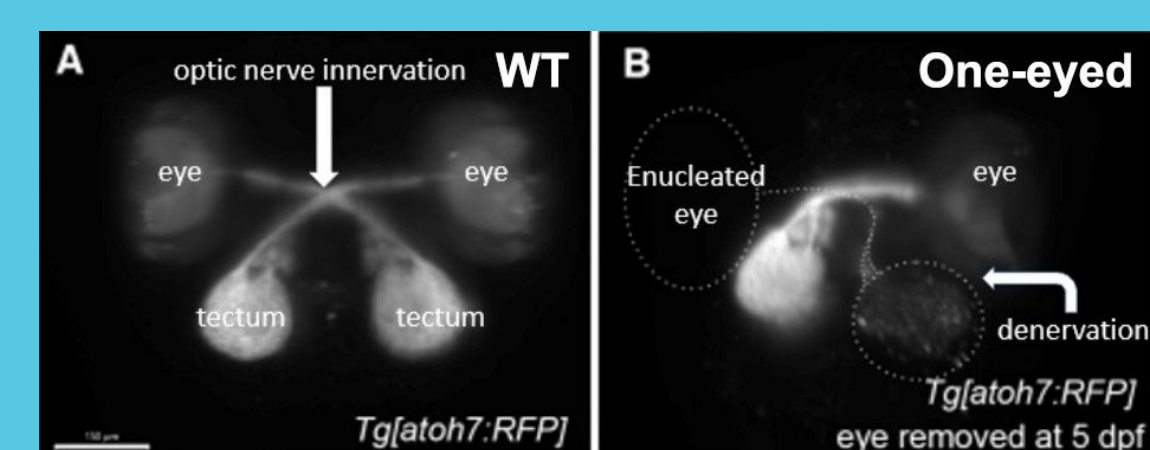
Approach

Paired transgenic fish with fluorescent markers for microglia (mpeg:mCherry)

Injected embryos at 1-cell stage with an apoptosis marker (Annexin A5:GFP)

Surgically removed one eye at 5 days old (dpf)

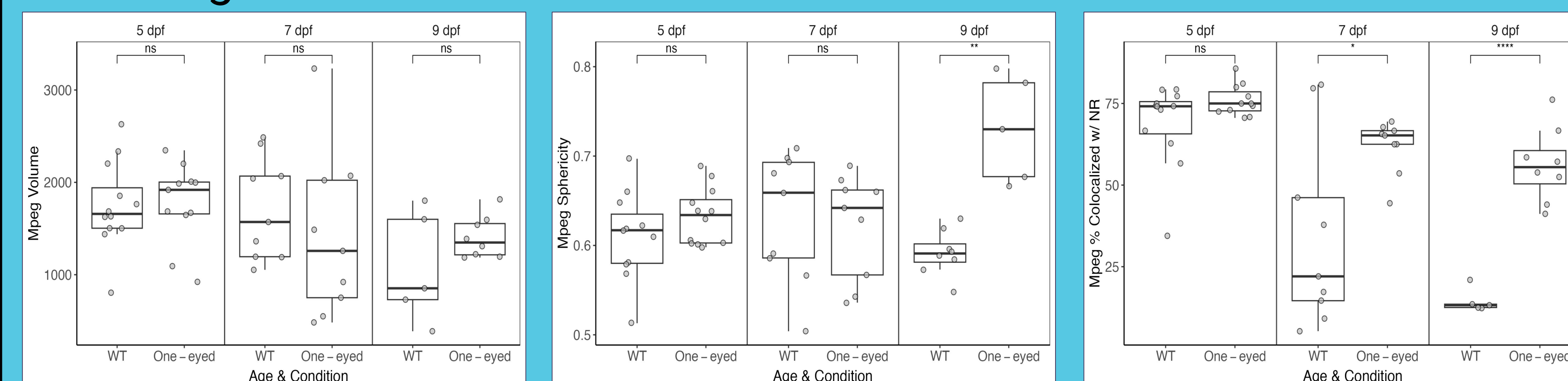
Live-cell time-lapsed larvae between 5 - 9 dpf



Eye removal leads to optic nerve degeneration (denervation)

Neutral Red - Mpeg

- Are microglia of one-eyed larvae more phagocytic?
- Phagocytic compartments stained with neutral red colocalize with microglia (mpeg)
- Microglia volume stays constant
- Increased microglia sphericity in one-eyed larvae at 9dpf indicates active phagocytosis and/or activation state changes



Microglia (mpeg:GFP) in green and phagocytic dots (Neutral Red) in magenta. Images acquired with fixed tissue confocal microscopy and quantified in Imaris 10.1. Box plots across ages and treatments; Unpaired t-test significance values: * $p \leq 0.05$, ** $p \leq 0.01$, **** $p \leq 0.0001$.

Conclusion

- Optic nerve denervation promotes microglia phagocytosis at 7 and 9 dpf
- Microglia display dynamic interactions with apoptotic cells and debris, including phagocytosis
- Microglia switch from homeostatic to activated
- Future directions involve imaging calcium flux, clonal cell death, and downstream projections in relation to denervation
- How can we better identify apoptotic versus non-apoptotic Annexin+ cells in time lapse images?

Acknowledgements

Thank you to Kara Cervený and the rest of the lab members. Funding support provided by the Arnold and Mabel Beckman Foundation, Helen Stafford Summer Research Fellowship Fund, and the Paul K. Richter & Evalyn Elizabeth Cook Richter Memorial Fund.